
ICPC North America Regional

2023 NCNA (Terse) Solutions

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8pm CT, Tuesday, November 7, 2023

Statistics

- <https://ncna23.kattis.com/contests/ncna23>
 - November 4, 2023 from 12pm CT to 5pm CT
 - **12** problems
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- **1346** submissions
 - **129** teams
 - **375** contestants

Difficulty

- **C:** 101/273
- **E:** 81/332
- **F:** 67/133
- **H:** 29/55
- **L:** 15/105
- **D:** 10/49

- **G:** 8/26
- **A:** 4/10
- **B:** 3/157
- **J:** 1/14
- **I:** 0/34
- **K:** 0/101

C. Anti-Palindrome (101/273)

- **linear scan**

- Remove punctuation
- Make every alphabets either lowercase or uppercase
- Check if the string contains a length 2 or 3 palindrome

E. Cuckoo! Cuckoo! (81/332)

- **bruteforce**

- First, increment the given time to end in 0, 15, 30, 45 minute mark
- Keep incrementing the time by 15 minutes until the total counts hit N
- Output the current time once the total count is reached

F. Cyclical Periods (67/133)

- **linear scan + custom sorting**

- For each character that appears in the sequence, save the difference between the indices of the second occurrence and the first occurrence (period) and the index of the first occurrence
- Follow the tie-break rule from the output section and output the answer
- Shorter alternative: You may just keep track of the first character you find that has repeated at each length (or the max length so far).

H. Modified Gray Code (29/55)

- ad hoc

- Convert the given number to binary in string s
 - Add 1 or 0 to the end of s depending on the parity of the bit count
 - Rectify the length of s to be 10 by inserting 0 in the front, as needed
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- Shorter alternative: Convince yourself that the value will never decrease when going from n^{th} to $(n+1)^{\text{st}}$. Then, just count up and find first number that has even bit difference with current number!

L. Scrabble Flash (15/105)

- **bruteforce**

- Pre-calculate the cost of i^{th} word and j^{th} words for any pair (i,j)
- Since N is at most 10, we may try all at most $10!$ permutations and keep track of the running maximum of $(U - \text{cost}[p[0]][p[1]] - \dots - \text{cost}[p[n-2]][p[n-1]])$ for a given permutation p .

D. Bombardment (10/49)

- **sorting**

- Sort x
- Keep simulating the process until no x is left while keeping track of the best hits
- Output the best hits saved

G. Gas Station (8/26)

- ad hoc

- Careful implementation problem

A. A Complex Problem (4/10)

- **strongly connected component (scc) + dp with memoization on directed acyclic graph [DAG] (condensation graph)**
- First, read in the directed graph
- Run strongly connected component algorithm (e.g., Kosaraju or Tarjan) to get a condensation graph (directed acyclic graph [DAG])
- Do dynamic programming with memoization on DAG

B. Add or Multiply (3/157)

- **data structure**

- Use segment tree with a state on each node
- Each node (state) keeps track of total, left, and right sums, number of terms, and whether product is used
- Care must be taken to merge the nodes in the segment tree

J. Ribbon Road (1/14)

- geometry

- Given the polygon P , it is sufficient to know how to check whether a point is inside, outside, or on the boundary of P .
- This is a well known geometry problem that can be solved using a cross product.

I. Reapportionment (0/34)

- **bitmask dp with memoization**

- If total sum is not divisible by W , output no immediately
- Pre-compute all the regions that have the sum equal to total / W
- We can do bitmask dp with memoization keeping track of the bitmask of towns covered and total number of wards processed
- If we hit the terminating condition of full bitmask and W wards processed, we may output yes and exit immediately; otherwise, output no

K. Scientific Grading (0/101)

- ad hoc (time waster - numerical analysis)
- Careful implementation problem
- Custom implementation of the functions may be needed;

Questions and Answers

- Any questions?

THANK YOU

